Experiment 2

Electrical Circuits

SAFETY

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances** attempt to repair any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

POWER SUPPLIES

In this lab we will be using a variable power supply. For all sections except "voltmeters" leave the power supply set to **6V**. Be sure to **turn off** the power supply when not in use.

Outline of Laboratory Exercise

In this lab you will be testing the basic ideas of electrical circuits. Resistors are the most common component in electrical circuits and therefore it is important to know how we can exploit their properties to make electrical circuits act as we like.

For a quick review on electrical circuits visit this site:

http://hyperphysics.phy-astr.gsu.edu/hbase/electric/ecircon.html and click on the links. The link Electric power relationship is useful for this lab.

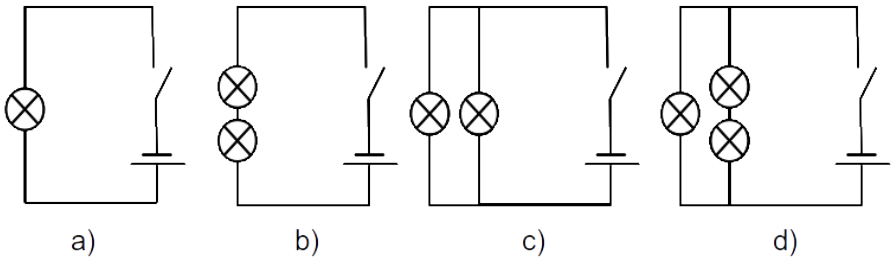
✓ ***Pre-lab exercises****: Read the laboratory exercise, then complete the online prelab questions. If you are unfamiliar with symbols used for electrical components in circuit diagrams, you should also read the relevant section in the Lab Appendices (Section D).*

Learning Goals

- To follow a circuit diagram to successfully wire and test the circuit.
- To troubleshoot and problem solve an incorrectly wired circuit and ensure it is re-wired correctly.
- To develop an understanding of how parallel and series circuits affect the power, current and voltage through a load.

Part A *Power Dissipation in Light Bulbs*

Firstly you will be predicting the brightness of the light bulbs when arranged in the following circuits. The brightness of an incandescent bulb is related to the power dissipated in it.



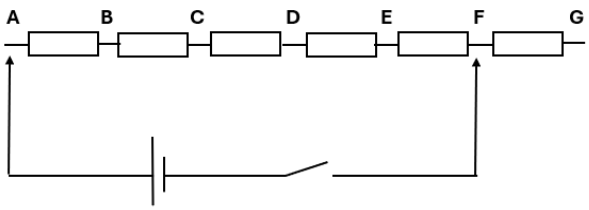
- Make a table of the predicted brightness of the bulbs. Use the bulb in circuit a) as the brightness standard and compare to this.
- Wire up the circuits. Do your predictions match your observations? Explain.

Part B *Ohm's law & **Voltage Dividers***

A **voltage divider** or **potential divider** is a circuit designed to divide a source of voltage ('potential difference') into sources of lower voltage.

In front of you there will be a chain of resistors in series all of the same value (nominally). Work out the value of these resistors using the colour chart in the lab book appendix. You will now be using this circuit to investigate voltage dividers. Voltage dividers work strictly on Ohm's law which states V = IR. For a specified supply voltage, you can predict the potential difference over a given (set of) resistor(s). Recall that the voltage across each individual resistive element in a circuit, and the total current through it, depends on the supply voltate *and* the other resistances in the circuit. We will use this fact to determine the average resistance of each resistor in a chain of nominally identical resistors.

In order to measure the current in the circuit below, consider where you will put an ammeter. Discuss with your demonstrator if you are unsure.



Gradually decrease the total resistance across the power supply, one resistor at a time until only one resistor remains, measuring the total current through the circuit at each stage, and plot the inverse of the current vs number of resistors.

Determine the value of the resistance of the resistors and compare to the one obtained from colour chart.

Now wire up the circuit so that **all** of the resistors are connected in series to the power supply (without the ammeter). Consider how you will connect a voltmeter in order to measure the potential differences in the table below. Discuss with your demonstrator if you are unsure. Draw a circuit diagram in your notebooks.

Measure the potential differences across the various terminals. Comment on your results and compare to your predicted values. If you have time, you may choose to create a plot to help with your analysis.

Terminals	Resistance (ohms)	Predicted Potential Difference (V)	Measured Potential Difference (V)
AA			
AB			
AD			
BF			
AG		6.0	

Part C Voltmeters

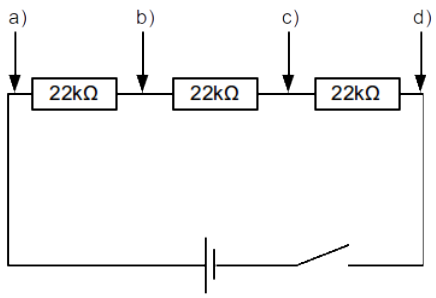
Note: For this section you can now change the voltage supply to 12V.

So far you have been investigating circuits with resistor of values much smaller than that in the voltmeter itself. Voltmeters work by using a galvanometer (device which measures current) and a known resistance to measure the voltage passing through it (Recall V=IR). By using a resistor of large value we can make a parallel circuit whose resistance is mostly unaffected by the resistance of the voltmeter. This can be seen using the parallel resistor equivalence relation.

$$\frac{1}{R_{eq}} = \frac{1}{R_{voltmeter}} + \frac{1}{R_{load}}$$

If the resistance of the voltmeter is very large the equivalent resistance is approximately equal to the load.

We will now investigate what happens when you measure the potential difference across a resistor of similar resistance to the voltmeter.



- Construct this circuit (the resistors are on your bench) and use the voltmeter to ensure that the power supply is maintaining $\mathbf{V_{AD}}$ at 12 V.

- From this value of $\mathbf{V_{AD}}$, use Ohm's Law to predict values for $\mathbf{V_{BD}}$, $\mathbf{V_{BC}}$ and $\mathbf{V_{AB}}$.
- Using the voltmeter, actually measure $\mathbf{V_{BD}}$, $\mathbf{V_{BC}}$ and $\mathbf{V_{AB}}$ using **both** the 0-50 V range and the 0-10 V range on the meter. If you are using a multimeter (as a voltmeter) ensure that it is in DC mode. Record your results.

Analysis

- Do the predicted and measured values agree? If not, can you explain why? Consider the **qualitative** effect of the voltmeter's resistance on the circuit.

Conclusion

- Summarise your main results.